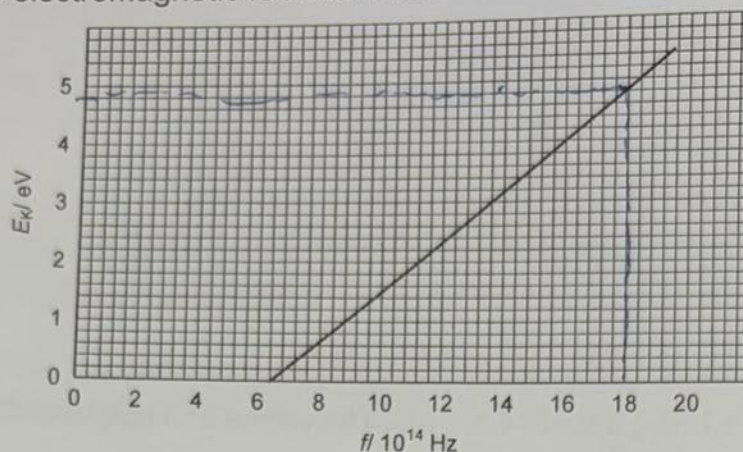


19.3 Discussion Questions

❖ Photoelectric Effect

- D1 The following graph shows how the maximum kinetic energy E_k of a photoelectron varies with the frequency f of the electromagnetic radiation that causes the emission of photoelectrons.



- (a) Use the graph to determine
- the threshold frequency for this material,
 - the maximum kinetic energy of photoelectrons from this material when it is illuminated with electromagnetic radiation of frequency 18.0×10^{14} Hz. Give your answer in joules.
- (b) Given $E_k = hf - \Phi$, where Φ is the minimum energy required to remove an electron from the metal surface. Use the graph to determine the Planck constant.

[N2000/3/6 modified]

❖ Wave-Particle Duality

- D2 A parallel beam of violet light of wavelength 4.5×10^{-7} m and intensity 700 W m^{-2} is incident normally on a surface.

- (a) Calculate
- the energy of a photon of violet light;
 - the number of photons incident per second on $1.0 \times 10^{-4} \text{ m}^2$.
- (b) (i) State the de Broglie relation for the momentum p of a particle in terms of its associated wavelength λ .
- (ii) Use the equation in (b)(i) to calculate the momentum of a photon of the violet light.
- (c) (i) Use your answers to (a) and (b) to calculate the change in momentum of photons incident on $1.0 \times 10^{-4} \text{ m}^2$ of the surface in one second. Assume that all photons are reflected by the surface.
- (ii) Suggest why the quantity you have calculated in (c)(i) is referred to as a 'radiation pressure'.

[N98/2/7]

❖ **Wavefunction of a Particle**

D4 A particle in an infinite square well has a wavefunction given by

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{4\pi x}{L}\right) & 0 \leq x \leq L \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Identify the quantum number n for the given wavefunction.
- (b) Sketch the probability density function $|\psi(x)|^2$, indicating nodes and maxima.
- (c) Hence determine the mean position of the particle.

D5 A particle in an infinite square well of width L is described by the wavefunction

$$\psi(x) = \begin{cases} A \cos\left(\frac{3\pi x}{L}\right) & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ 0 & \text{for other values of } x \end{cases}$$

- (a) What is the meaning of normalising a wavefunction?
- (b) Determine the normalization constant A .

D6 In a simple model of the hydrogen atom, the electron is allowed to move only along a straight line of length a . The associated wave motion must always show a standing wave pattern along this line, with nodes at the ends.

- From the condition that the standing waves must have nodes at the ends of the line of length a , find the allowed values of the momentum of the electron.
- Hence, show that the allowed values of E_n of the energy of the electron of mass m_e in this model of the hydrogen atom are given by $E_n = \frac{n^2 h^2}{8m_e a^2}$, $n = 1, 2, 3, \dots$

D7 A Cadmium Selenide (CdSe) quantum dot is a semiconductor nanoparticle that confines electrons in all three dimensions. For a simplified model, we can treat an electron in the dot as a particle in a one-dimensional infinite potential well of width L .

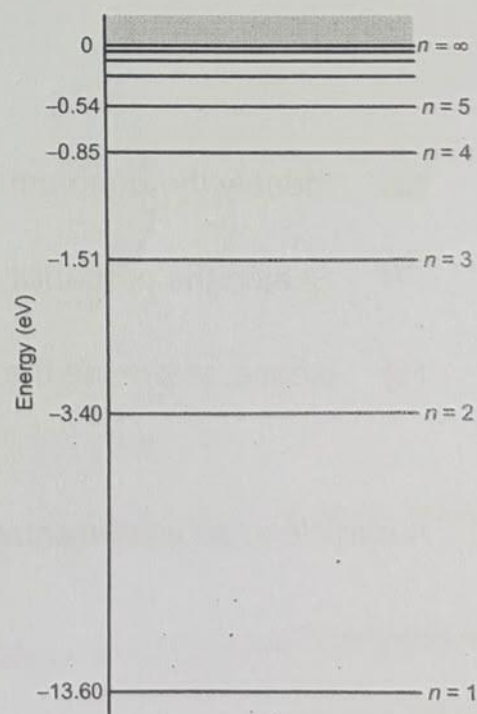
When an electron in this quantum dot drops from the first excited state ($n = 2$) to the ground state ($n = 1$), it emits a photon of red light with a wavelength of 694.3 nm.

- Given that the effective mass of an electron in CdSe is 0.13 times the mass of a free electron, calculate the width L of the quantum dot.
- A second CdSe quantum dot is manufactured with a smaller width than the first. Predict how the wavelength and the color of the emitted light would change.

❖ Line Spectra

D8 The diagram below represents some of the energy levels in a hydrogen atom.

- Use the diagram to explain what is meant by the terms
 - ground state,
 - excitation energies and
 - ionisation energy
- State what transitions may happen when a photon of light hits and interacts with the hydrogen atom (initially at its ground state), when its energy content is
 - 6.50 eV
 - 10.20 eV
 - 11.40 eV
 - 12.09 eV

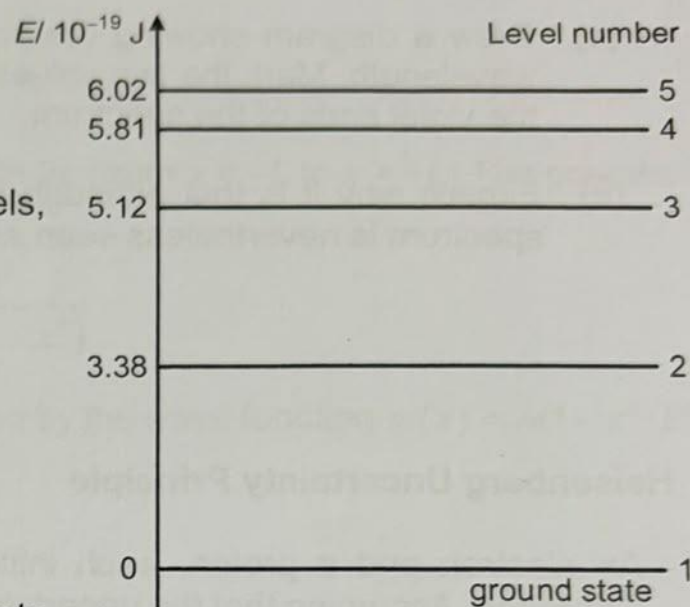


- State what may happen if an electron with the same energies as the photon hits and interacts with the hydrogen atom (initially at its ground state).

- (c) State what may happen if an electron with the same energies as the photon hits and interacts with the hydrogen atom (initially at its ground state).

- (i) 6.50 eV
- (ii) 10.20 eV
- (iii) 11.40 eV
- (iv) 12.09 eV

D9 The diagram is a simplified representation of the 5 lowest energy levels of the outermost electron in the sodium atom.



- (a) Considering transitions between only these levels,
 - (i) which spectral transition has the shortest wavelength (give your answer in terms of level numbers),
 - (ii) how many spectral emission lines might be produced by transitions among these levels?
- (b) If the sodium atoms are initially in the ground state, how many absorption lines might be detected?
- (c) Cool sodium vapour at low pressure is bombarded with electrons of kinetic energy E . Which transitions would you expect to observe if E has the value
 - (i) $3.0 \times 10^{-19} \text{ J}$
 - (ii) $4.0 \times 10^{-19} \text{ J}$
 - (iii) $5.5 \times 10^{-19} \text{ J}$
- (d) In practice, the highest level inferred from observation of single electron transitions in the sodium spectrum is $8.21 \times 10^{-19} \text{ J}$.
 - (i) Explain the significance of this value.
 - (ii) Calculate the range of potential differences which would accelerate bombarding electrons to produce spectral line emission, but no free electrons.
- (e) In fact, level 2 consists of a pair of closely spaced levels. Transitions from them give rise to sodium "D-lines" of wavelengths 589.0 nm and 589.6 nm.
 - (i) Calculate the energy difference between the two closely spaced levels.
 - (ii) Identify the other level involved from which the transitions took place.

- D10** The measured wavelengths, λ_m , of selected lines in the hydrogen spectrum are given empirically by

$$\frac{1}{\lambda_m} = R \left(\frac{1}{4} - \frac{1}{m^2} \right)$$

where R is a constant and has the value $1.097 \times 10^7 \text{ m}^{-1}$ and m is an integer taking the values 3, 4, 5, ...etc.

- (a) Calculate the value of the wavelength when $m = 4$.
- (b) Calculate the minimum wavelength given by this equation.
- (c) Draw a diagram showing the approximate positions of the lines on a horizontal axis of wavelength. Mark the two values you have already calculated and also mark the red and the violet ends of the spectrum.
- (d) Explain why it is that although there is an infinite number of lines in this spectrum, the spectrum is nevertheless seen as a line spectrum.

[J88/2/12 part]

❖ Heisenberg Uncertainty Principle

- D11** An electron and a proton, each initially at rest, are accelerated across the same potential difference. Assuming that the uncertainty in their position is given by their de Broglie wavelength, calculate the ratio of the uncertainty in their momentum.
- D12** What is the uncertainty in the location of a photon of wavelength 300 nm if this wavelength is known to an accuracy of one part in a million?
- D13** Consider a particle trapped in an infinite square well.
Explain why
- (a) the particle has definite energy but **not** definite momentum,
 - (b) the lowest possible energy of the particle cannot be zero.